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Weisfeiler and Lehman Go Paths: Learning Topological Features via Path Complexes

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MOTIVATION

Prior works on topological higher-order GNNs often depend on assumptions about sub-structures of graphs, such as cliques, cycles, and rings.

Our study presents a novel perspective by focusing on **simple paths**, a universal element in graphs, during the topological message-passing process.

LIFTING TRANSFORMATIONS

Given a simple graph $G = (\mathcal{V}, \mathcal{E})$ with a finite vertex set $\mathcal{V}$ and edge set $\mathcal{E}$, we can apply lifting transformation such as clique [2] cell [3], or path complex lifting.

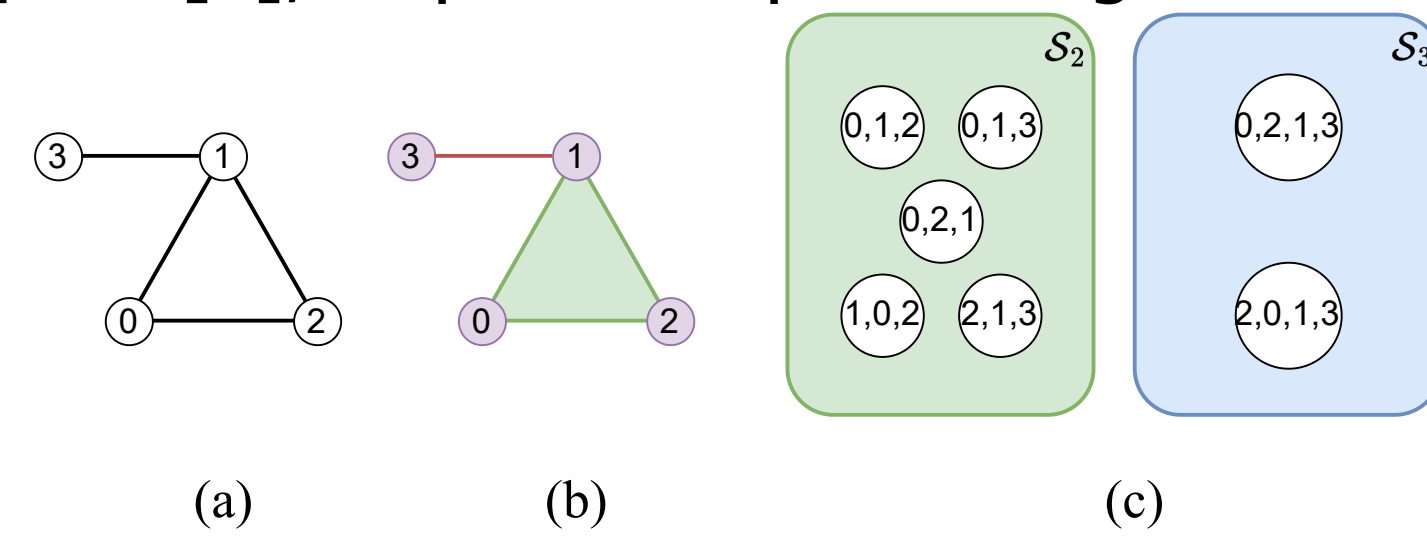


Figure 1. (a) Original graph; (b) Simplicial complex, which contains a 2-simplex, 4 1-simplices, and 4 0-simplices, arising from the original graph. Regular cell complex coincides with the simplicial complex in this case; (c) Simple path spaces $\mathcal{S}_2$ and $\mathcal{S}_3$ corresponding to the path complex arising from the original graph. Elementary paths of $\mathcal{S}_0$ and $\mathcal{S}_1$ are indeed 0-simplices (0-cells) and 1-simplices (1-cells) of the simplicial complex (regular cell complex).

TUDATASETS

Dataset	PROTEINS	NCI1	NCI109	IMDB-B
RWK	59.6 ± 0.1	> 3 days	N/A	N/A
GK (k=3)	71.4 ± 0.3	62.5 ± 0.3	62.4 ± 0.3	N/A
PK	73.7 ± 0.7	82.5 ± 0.5	N/A	N/A
WL Kernel	75.0 ± 3.1	86.0 ± 1.8	N/A	73.8 ± 3.9
DCNN	61.3 ± 1.6	56.6 ± 1.0	N/A	49.1 ± 1.4
DGCNN	75.5 ± 0.9	74.4 ± 0.5	N/A	70.0 ± 0.9
IGN	76.6 ± 5.5	74.3 ± 2.7	72.8 ± 1.5	72.0 ± 5.5
GIN	76.2 ± 2.8	82.7 ± 1.7	N/A	75.1 ± 5.1
PPGNs	77.2 ± 4.7	83.2 ± 1.1	82.2 ± 1.4	73.0 ± 5.8
Natural GN	71.7 ± 1.0	82.4 ± 1.3	N/A	73.5 ± 2.0
GSN	76.6 ± 5.0	83.5 ± 2.0	N/A	77.8 ± 3.3
pathGCN	80.4 ± 4.2	83.3 ± 1.3	N/A	N/A
PathNN	75.2 ± 3.9	82.3 ± 1.9	N/A	72.6 ± 3.3
SIN [†]	76.4 ± 3.3	82.7 ± 2.1	N/A	75.6 ± 3.2
CIN [†]	77.0 ± 4.3	83.6 ± 1.4	84.0 ± 1.6	75.6 ± 3.7
CAN	78.2 ± 2.0	84.5 ± 1.6	83.6 ± 1.2	N/A
CIN++	80.5 ± 3.9	85.3 ± 1.2	84.5 ± 2.4	N/A
PIN (Ours)	78.8 ± 4.4	85.1 ± 1.5	84.0 ± 1.5	76.6 ± 2.9

SR GRAPHS

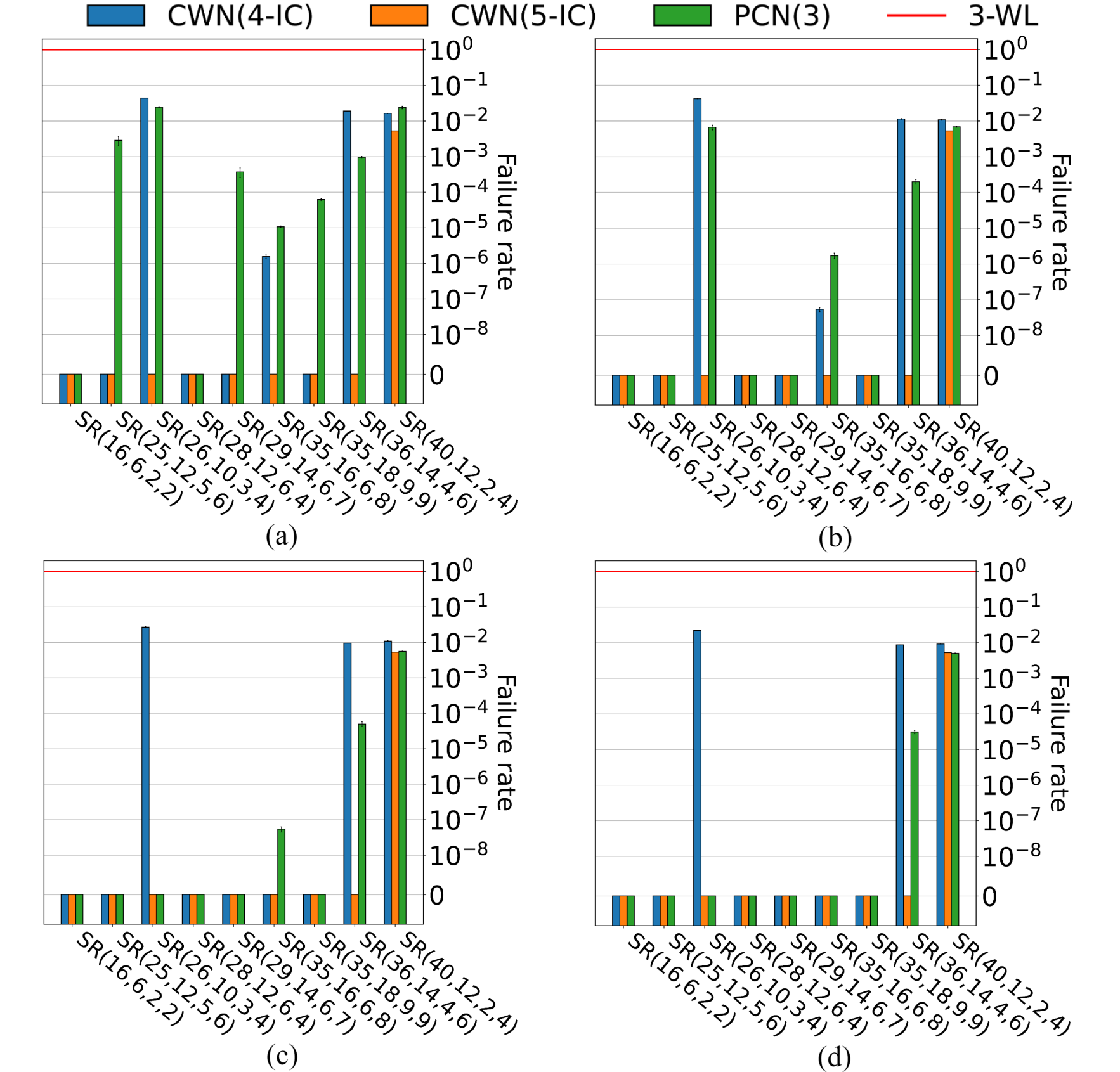


Figure 3. Failure rate comparison between CWN and PCN on SRG Families over 10 different seeds. (a) 3 MP layers. (b) 4 MP layers. (c) 5 MP layers. (d) 6 MP layers.

PATH COMPLEXES

Definition 1. (Grigor'yan et al. [1]) Given a finite non-empty set V whose element is called vertex, an **elementary p-path** on set V is any sequence of vertices with length $p + 1$. Elementary p -path is denoted by $e_{i_0 \dots i_p}$.

Definition 2. (Grigor'yan et al. [1]) Given a finite non-empty set V , a **path complex** P is a non-empty collection of elementary paths such that for any sequence of vertices that belong to P , the truncated sequences, in which either the first vertex or the last vertex is removed, are also included in P .

THEORETICAL RESULTS

Proposition 3. For any two path complexes X and Y , if $\sigma \in X$ and $\tau \in Y$ have different boundary sizes $|\mathcal{B}(\sigma)| \neq |\mathcal{B}(\tau)|$, their colorings are different $c_\sigma^{X,t} \neq c_\tau^{Y,t}$ for $t > 0$.

Theorem 4. PWL is at least as powerful as SWL [2] at distinguishing non-isomorphic graphs.

Theorem 5. WL is at least as powerful as CWL(k -IC) [3] at distinguishing non-isomorphic graphs.

Corollary 6. PWL is strictly more powerful than WL at distinguishing non-isomorphic graphs.

Corollary 7. PWL is not less powerful than 3-WL at distinguishing non-isomorphic graphs.

ILLUSTRATION

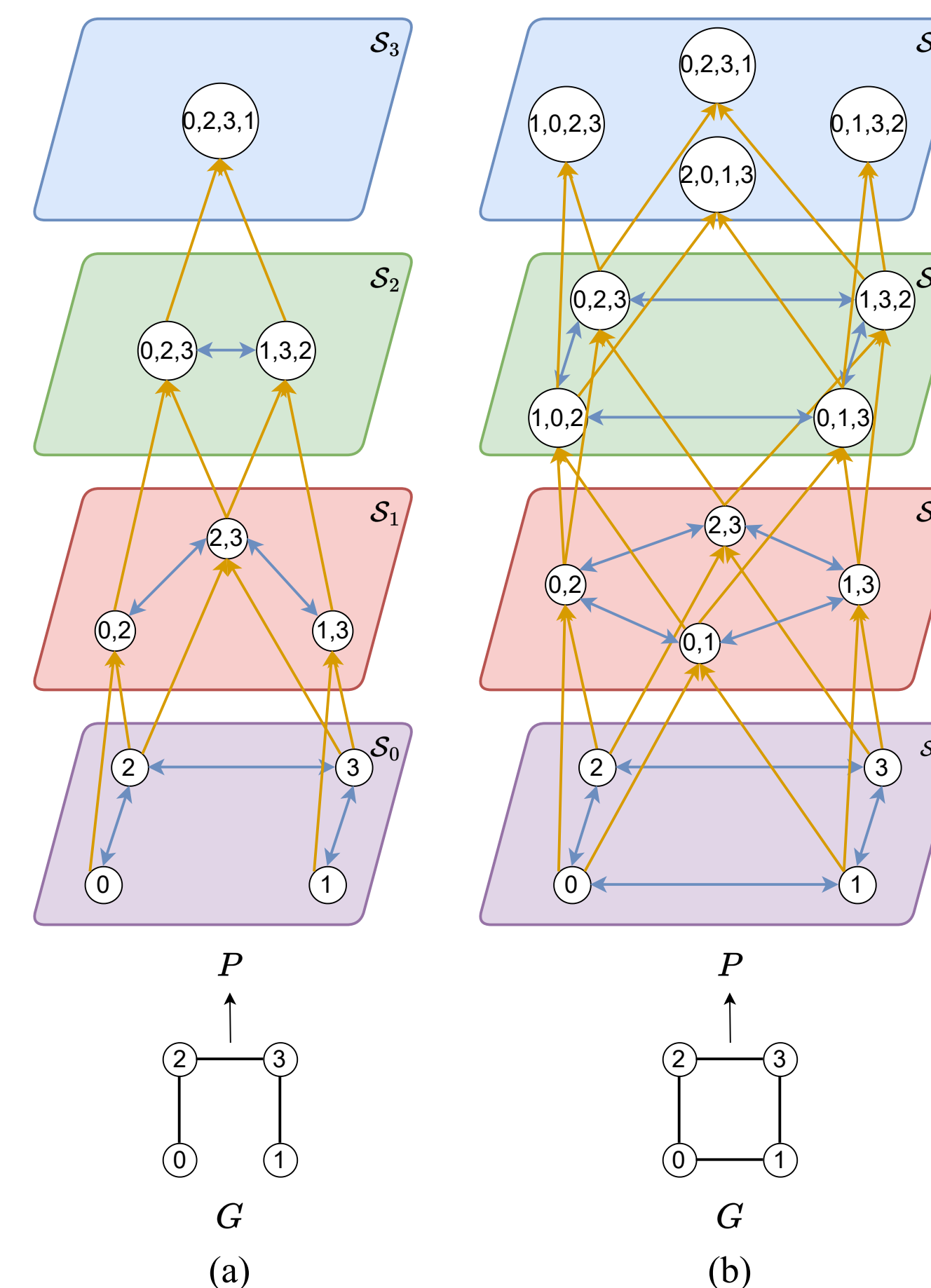


Figure 2. Examples of path complexes arising from graphs.

ZINC & OGBG-MOLHIV

Dataset	ZINC		OGBG-MOLHIV	
	No Edge Feat.	W/ Edge Feat.	Test ROC-AUC	Val. ROC-AUC
GCN	0.469 ± 0.002	N/A	N/A	N/A
GAT	0.463 ± 0.002	N/A	N/A	N/A
GatedGCN	0.422 ± 0.006	0.363 ± 0.009	N/A	N/A
GIN	0.408 ± 0.008	0.252 ± 0.014	77.07 ± 1.49	84.79 ± 0.68
PNA	0.320 ± 0.032	0.188 ± 0.004	79.05 ± 1.32	85.19 ± 0.99
DGN	0.219 ± 0.010	0.168 ± 0.003	79.70 ± 0.97	84.70 ± 0.47
HIMP	N/A	0.151 ± 0.006	78.80 ± 0.82	N/A
GSN	0.140 ± 0.006	0.115 ± 0.012	77.99 ± 1.00	86.58 ± 0.84
PathNN	N/A	0.090 ± 0.004	79.17 ± 1.09	N/A
CIN [†]	0.115 ± 0.003	0.079 ± 0.006	80.94 ± 0.57	N/A
CIN++	N/A	0.077 ± 0.004	80.63 ± 0.94	N/A
PIN (Ours)	0.139 ± 0.004	0.096 ± 0.006	79.44 ± 1.40	82.41 ± 0.96

REFERENCES

- [1] Grigor'yan, A.; Lin, Y.; Muranov, Y.; and Yau, S.-T. 2013. Homologies of path complexes and digraphs. arXiv: 1207.2834 [math].
- [2] Bodnar, C.; Frasca, F.; Otter, N.; Wang, Y.; Li, P.; and Bronstein, M. 2021b. Weisfeiler and Lehman Go Topological: Message Passing Simplicial Networks. In Meila, M., and Zhang, T., eds., Proceedings of the 38th International Conference on Machine Learning, volume 139 of Proceedings of Machine Learning Research, 1026–1037. PMLR.
- [3] Bodnar, C.; Frasca, F.; Otter, N.; Wang, Y.; Li, P.; Montufar, G. F.; and Bronstein, M. 2021a. Weisfeiler and Lehman Go Cellular: CW Networks. In Ranzato, M.; Beygelzimer, A.; Dauphin, Y.; Liang, P.; and Vaughan, J. W., eds., Advances in Neural Information Processing Systems, volume 34, 2625–2640. Curran Associates, Inc.